

Homework #2

Answer the following questions. Use your own words while answering the questions. Write your answers on a paper and submit your solutions to me during the class hour on **May 06, 2019**.

1. Assume the following JavaScript program was interpreted using static-scoping rules. What value of `x` is displayed in function `sub1`? Under dynamic-scoping rules, what value of `x` is displayed in function `sub1`?

```
var x;
function sub1() {
    document.write("x= " + x + "<br />");
}
function sub2() {
    var x;
    x = 20;
    sub1();
}
x = 10;
sub2();
```

2. Explain the problems given in the below C code segment when it is executed.

```
int Sum[100], *p1, *p2, *p3;
p1 = (int *)malloc(sizeof(int)*200);
p2 = p1;
free(p2);
p3 = (int *)malloc(sizeof(int)*300);
p3 = Sum;
```

3. Assume the following rules of associativity and precedence are defined for expressions:

Precedence	Highest	*, /, not +, -, &, mod Unary - ==, !=, <, <=, >, >=
	Lowest	and or, xor
Associativity	Left-to-right	

Show the order of evaluation of the following expressions by parenthesizing all subexpressions and placing a superscript or subscript on the right parenthesis to indicate the order.

`a mod b * 7 + d`

`a * b + x == c + d mod 5 and x != y`

`-a + b or c == d & f`

4. Consider the following C program:

```
int fun(int *i){
    *i += 2;
    return 10;
}
void main(){
    int x = 3;
    x = x + fun(&x) + x + fun(&x);
}
```

What is the value of x after the assignment statement in main, assuming

- a) operands are evaluated left to right.
- b) operands are evaluated right to left.